

Soowan Yang

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EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Feb. 2027 – (Incoming)
M.S.–Ph.D. Integrated Program, Kim Jaechul Graduate School of AI
Advisor: **Prof. Donghyun Kim** (DARoS Lab)

Daegu Gyeongbuk Institute of Science and Technology (DGIST) Feb. 2020 – Feb. 2027 (expected)
B.S. in Mechanical Engineering
GPA: 3.96 / 4.30 (magna cum laude)

University of California, Berkeley Jun. 2024 – Aug. 2024
Exchange Student (DGIST FGLP Program)

- Completed coursework in applied mathematics and engineering

RESEARCH EXPERIENCE

KAIST (DRCD Lab) Feb. 2026 – Aug. 2026
Undergraduate Researcher (Advisor: Prof. Haewon Park) Daejeon, South Korea

- Retargeted SMGDiff Soccer-X human soccer motions to a 27-DoF KAIST humanoid via GMR, building a feasible motion-clip library
- Built a two-layer controller: an MCTS planner selecting macro-actions (dribble angle / skill) over a motion-matching (kNN) clip selector for natural execution
- Achieved real-time (30 Hz) dynamic obstacle avoidance, dribbling past static cones and moving defenders via learned soccer tricks

University of Massachusetts Amherst (DARoS Lab) Jun. 2025 – Dec. 2025
Visiting Researcher (Advisor: Prof. Donghyun Kim) Amherst, MA, USA

- Worked on assistive robotics for blind and low-vision navigation (GuideNav project)
- Collected and processed tactile walking surface indicator (TWSI) data from real-world environments
- Built and curated large-scale real-world + synthetic datasets for perception (GuideTWSI, NavAble)

DGIST (MCL Lab) Mar. 2024 – Jun. 2025
Undergraduate Researcher (Advisor: Prof. Sehoon Oh) Daegu, South Korea

- Participated in robotics and AI research projects
- Worked on perception and dataset-related tasks for robotics applications

PUBLICATIONS

- **NavAble: A Large-Scale Dataset and Synthetic Data Generation Pipeline for Blind Navigation**
Hochul Hwang[†], Jahir Sadik Monon[†], **Soowan Yang**, Shiven Umeshbhai Patel, Anh Nhat Hong Nguyen, Kien Nguyen, Keshav Garg, Dylan Gage, Duretti Hordofaa, Anshu Anjna, Eshed Ohn-Bar, Donghyun Kim († equal contribution)
Conference on Neural Information Processing Systems (NeurIPS), Evaluations & Datasets Track, 2026
- **GuideTWSI: A Diverse Tactile Walking Surface Indicator Dataset from Synthetic and Real-World Images for Blind and Low-Vision Navigation**
Hochul Hwang, **Soowan Yang**, Anh N. H. Nguyen, Parth Goel, Krisha Adhikari, Sunghoon I. Lee, Joydeep Biswas, Nicholas A. Giudice, Donghyun Kim
IEEE International Conference on Robotics and Automation (ICRA), 2026
🏆 **Best Paper Award Finalist — Field and Service Robotics**
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- **GuideNav: User-Informed Development of a Vision-Only Robotic Navigation Assistant for Blind Travelers**
Hochul Hwang, **Soowan Yang**, Jahir Sadik Monon, Nicholas A. Giudice, Sunghoon Ivan Lee, Joydeep Biswas, Donghyun Kim
ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2026
 **Systems Paper Honourable Mention**
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HONORS & AWARDS

- **DGIST President's Award** — Top Honor, Student Category, DGIST Foundation Day Sep. 2026
- **Best Paper Award Finalist** (Field and Service Robotics), ICRA 2026 2026
- **Systems Paper Honourable Mention**, ACM/IEEE HRI 2026 2026
- **Magna cum laude**, B.S. in Mechanical Engineering, DGIST 2027 (expected)

PROJECTS

Humanoid Soccer: MCTS Planning with Motion Matching

- Retargeted SMGDiff Soccer-X motions to a KAIST 27-DoF humanoid with GMR, building a clip library for dribbling and tricks
- Designed a two-layer planner: receding-horizon MCTS (UCB tree search, MPC-style replanning) over a motion-matching (kNN) layer for smooth, feasible motion
- Enabled dynamic obstacle breakthrough — dribbling past static cones and moving defenders using soccer tricks (step-over, Marseille turn) with a ball-shielding mechanism
- Reached real-time control at 30 Hz (~ 250 simulations/step), validated across multiple field scenarios in MuJoCo

Guide-Dog Robot Perception and Navigation

- Built a vision-based perception system for tactile paving detection (YOLO, SAM2)
- Achieved 96% accuracy in real-world deployment (Boston subway)
- Conducted user studies with visually impaired participants and deployed on Unitree Go2
- Co-authored papers at NeurIPS 2026, HRI 2026 ( **Honourable Mention**) and ICRA 2026 ( **Best Paper Finalist**)

Assistive Mechatronics Device

- Developed an assistive robotic device for grasp augmentation
- Received top evaluation at DGIST poster session

Manipulator Control and Simulation

- Developed real-time control and MuJoCo simulation for manipulators
- Resolved instability from discrete-time dynamics mismatch, achieving stable control

SKILLS

Programming: Python, C++

Robotics: ROS2, Isaac Sim, MuJoCo

Planning & Learning: MCTS, Motion Matching, Reinforcement Learning

Vision: YOLO, SAM, OpenCV

Tools: Git, Linux, Docker